

Sara Klaasen

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Education

PhD	ETH Zurich , Geophysics • ETH medal	Zurich, Switzerland 2020 – 2025
Msc	ETH Zurich , Geophysics	Zurich, Switzerland 2018 – 2020
Bsc	UCR , Liberal Arts and Sciences • Magna cum Laude • Major in Earth sciences, biology, mathematics and physics	Middelburg, Netherlands 2014 – 2017

Experience

Australian National University , Postdoctoral researcher Researching DAS projects in New Zealand. • Imaging Taupo Volcano.	Canberra, Australia 2026 – present 1 year
ETH Zurich , Postdoctoral researcher	Zurich, Switzerland 2025 – 2026 1 year
SRO , Admin Online administration	Haarlem, Netherlands 2017 – 2017 1 year
PZC , Online editor • Online content management of the provincial newspaper • Writing and editing news articles	Vlissingen, Netherlands 2015 – 2017 2 years

Awards

ETH Medal The ETH Medal is given to outstanding doctoral theses (2k CHF). ETH Zurich	2026
OSPA Outstanding student presentation award Seismological Society of America Photonics Meeting	2024

Grants

SNF postdoc.mobility grant Grant to fund a 2-year postdoc position (136k CHF) Swiss National Science Foundation	2025
Women-in-STEM grant Grant for peer-mentoring group (2k CHF). Fix the Leaky Pipeline program, ETH Zürich	2021
VSBFonds grant Grant to support a Masters programme abroad (10k EUR). VSBFonds	2018

Research collaborations

ARC Centre of Excellence in Optical Microcombs for Breakthrough Science (COMBS)

2026 – present

Associate investigator

www.combs.org.au

Seismicity on Venus: Prediction and Detection

2022 – 2024

International Space Science Institute (ISSI) international team.

ECR member

teams.issibern.ch/seismicityonvenus

Teaching experience

Data Analysis and Visualisation with Python in Earth Sciences

2022 – 2025

Bachelor course at ETH Zürich

Teaching assistant

Geophysical field lab

2021 – 2023

Bachelor fieldcourse at ETH Zürich

Teaching assistant

Dynamische Erde I

2020 – 2023

Annual ten minute lectures about current geophysical applications during the Bachelor course at ETH Zürich.

Guest lecturer

Books

Fiber optic sensing for volcano monitoring and imaging volcanic processes

2025

Book chapter in Modern Volcano Monitoring

Philippe Jousset, Gilda Currenti, Shane Murphy, Eva Eibl, Corentin Caudron, Lise Retailleau, Christopher Wollin, Sara Klaasen, Takeshi Nishimura, Andreas Fichtner, Zack Spica, Arnaud Lemarchand, Charlotte Krawczyk

doi.org/10.1007/978-3-031-86841-2_17 (Springer)

The leading edge of volcanology: Advances and innovations in monitoring and interpreting volcanic activity

2027

Book chapter submitted to "The encyclopedia of Volcanoes, Third Edition". Chapter has been accepted.

Poland, Michael P and de Zeeuw-van Dalfsen, Elske and Aizawa, Koki and Klaasen, Sara and D'Arcy, Fiona and Nicholson, Emma J and Anderson, Kye
(Elsevier)

Publications

Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano

2026

Sara Klaasen, Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner

doi.org/10.26443/seismica.v5i1.1845 (Seismica)

Seismic constraints on glacier density

2025

Ariane Lanteri, Scott Keating, Lars Gebraad, Sara Klaasen, Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner

doi.org/10.1038/s41598-025-26440-z (Scientific Reports)

- The Seismogenic Thickness of Venus** 2025
Julia Maia, Ana-Catalina Plesa, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, Sara Klaasen, Robert Herrick, Raphael Garcia
doi.org/10.1038/s41598-025-26440-z (Journal of Geophysical Research: Planets)
- Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibreoptic-seismometer networks** 2025
Thomas Hudson, Sara Klaasen, Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner
doi.org/10.1093/gji/ggae459 (Geophysical Journal International)
- An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology** 2024
Andreas Fichtner, Fabian Walter, Patrick Paitz, Sara Klaasen, Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel
dx.doi.org/10.1016/j.eq.s.2024.09.006 (Earthquake Sciences)
- Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers** 2024
Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, Sara Klaasen, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina Plesa, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli
doi.org/10.1029/2024EA003670 (Earth and Space Science)
- Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables** 2024
Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, Sara Klaasen, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti
doi.org/10.1093/gji/ggae364 (Geophysical Journal International)
- Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments** 2024
Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, Sara Klaasen, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu
doi.org/10.4430/bgo00458 (Bulletin of geophysics and oceanography)
- Challenges in submarine fiber-optic earthquake monitoring** 2024
Jonas Igel, Sara Klaasen, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner
doi.org/10.1029/2024JB029556 (Journal of Geophysical Research: Solid Earth)
- Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland** 2023
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner
doi.org/10.30909/vol.06.02.301311 (Volcanica)
- Fiber-optic observation of volcanic tremor through floating ice sheet resonance** 2022
Andreas Fichtner, Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir
doi.org/10.1785/0320220010 (The Seismic Record)

Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia	2021
Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner doi.org/10.1029/2021JB022358 (Journal of Geophysical Research: Solid Earth)	
Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing	2021
Solvi Thrastarson, Robert Torfason, Sara Klaasen, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner doi.org/10.1002/essoar.10509693.1 (ESS Open Archive)	
Talks (invited)	
Keck observatory workshop	2026
SSA optical seismology and the next era of seismic sensing, Kona (Workshop speaker)	
Session: Frontiers in Fiber-Optics Sensing in Seismology	2025
IASPEI-IAGA, Lisbon (Invited speaker)	
ECR Keynote talk	2025
IAVCEI, Geneva (Invited keynote speaker)	
Seminar	2025
Stanford University, San Francisco (Seminar speaker)	
Seminar	2024
University of Washington, online (Seminar speaker)	
Session: An innovative photonic vision of volcanoes and geothermal systems	2024
SSA Photonic Seismology, Vancouver (Invited speaker)	
Colloquium	2023
CSIRO, Perth (Seminar speaker)	
Geophysics Weekly Meeting	2023
Australian National University, Canberra (Seminar speaker)	
Seminar	2023
University of Canterbury, Christchurch (Seminar speaker)	
Workshop on Distributed Acoustic Sensing	2021
AGU, online (Workshop speaker)	

Service

Convening conference sessions
IAVCEI, AGU

Peer-review

Seismica, SRL

Languages

Dutch

native

English

C2

German

B2

French

B1

Conference contributions (selected)

Modelling months of deformation during the Sundhnúkur eruptions on the Reykjanes Peninsula, Iceland	2026
Sara Klaasen, Daniel Bowden, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Jack Maxwell, Meghan Miller, Andreas Fichtner (SSA optical seismology and the next era of seismic sensing)	
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece	2025
Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (IAVCEI General Meeting)	
Full-Waveform Inversion of DAS data on a subglacial volcano: Grímsvötn, Iceland	2025
Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner (IAVCEI General Meeting)	
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece	2025
Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (SSA General Meeting)	
Moment Tensor inversion with Full-Waveform Inversion and Distributed Acoustic Sensing on a subglacial volcano: Grímsvötn, Iceland	2024
Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU Copernicus Meetings)	
Submarine volcano monitoring with fiber-optic cables at Santorini	2023
Sara Klaasen, Paraskevi Nomikou, Konstantinos Karantzas, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	
Going towards Full Waveform Inversion (FWI) with Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland	2023
Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	

Volcano monitoring and modelling using Distributed Acoustic Sensing (DAS) on a sub-glacial volcano: Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Sebastian Noe, Andreas Fichtner (IUGG General Assembly)	2023
Monitoring glacially covered volcanoes with Fibre-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (IAVCEI General Meeting)	2023
Volcanic Event Characterization with Fiber-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (AGU Annual Meeting)	2022
Fibre-Optic Sensing for Volcano Monitoring on Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU General Assembly)	2022
Distributed Acoustic Sensing on Volcanoes: Grímsvötn and Fagradalsfjall, Iceland Sara Klaasen, Solvi Thrastarson, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Patrick Paitz, Andreas Fichtner (AGU Annual Meeting)	2021
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner (SSA General Meeting)	2021
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner (EGU General Assembly)	2021
Distributed Acoustic Sensing for the Exploration of the Mount Meager Volcanic Complex, British Columbia, Canada Sara Klaasen, Andreas Fichtner, Patrick Paitz, Jan Dettmer (AGU Annual Meeting)	2020